

JENNY ZHANG

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TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, C++, C, C#, Bash, SQL, PHP, HTML

Tools / Frameworks: Git, Docker, AWS, React, Vue, Node, Flask, Laravel, MySQL, MongoDB

EDUCATION

University of Waterloo

Waterloo, ON

Bachelor of Computer Science Co-op Program

September 2021 - May 2026

- **Specializations and Minors:** Artificial Intelligence (AI) Specialization
- **Relevant courses:** Artificial Intelligence, Neural Networks, Machine Learning, Data Structures & Algorithms, Computer Networks, Object-Oriented Programming, Operating Systems, Compilers, Statistics, Linear Algebra

EXPERIENCE

Software Engineer Intern | Rippling, San Francisco

May 2025 – August 2025

- Incoming Software Engineer Intern at Rippling

Software Developer Intern | ArenaX Labs, Toronto

September 2024 – December 2024

- Developed attack-combo mechanics for AI-powered characters using **finite state machines**; implemented action sequences and leveraged **class inheritance** to customize transition functions across states
- Refactored game UI in **React** and **TypeScript**, leveraging **Jotai** for state management to achieve **30%** faster data persistence across component hierarchies, reducing render time and improving gameplay smoothness

Fullstack Developer Intern | Knak, Ottawa

January 2024 – August 2024

- Streamlined marketing processes by **95%** for enterprise companies through developing Knak's editor platform using **Vue**, **PHP**, **AWS Lambda**, and **Docker**; allowing clients to create email and landing pages within minutes
- Optimized data retrieval from the **MySQL** database to dynamically display translation workflow options on the frontend based on customer permissions, enhancing user experience in Knak's AI-driven text translation tool
- Engineered optimized **MySQL** queries in **PHP**, boosting query performance by **30%** and enabling seamless client data migration within seconds during feature deployments

Game Developer Intern | ZeMind Game Studios, Toronto

May 2023 – August 2023

- Built a virtual reality (VR) application in **Unity/C#** for Durham Region Victim Services (DRVS), providing three immersive experiences to help youth navigate exploitative situations; nominated for the Webby awards 2024
- Orchestrated the engineering of a dialogue and scene management system using an **observer design pattern**

Software Developer Intern | Royal Bank of Canada, Toronto

May 2022 – August 2022

- Designed and developed a **Flask** application to test **60%** of RBC's site actions with **Python**, **Pytest** and **Selenium**; adopted by team of 10 developers, reducing CI pipeline test workflow duration by **80%**

PROJECTS

🔗 **Landing Page Generator** | JavaScript, React, Node, MongoDB, Cohere LLM API, OpenAI API

- Link platform that provides a codeless editor to create personalized domains for users in under **2 minutes**

🔗 **Emulator for Game Boy** | C, CMake, SDL

- Game Boy CPU core emulator, effectively translating **C** code into the SM83 processor's instruction set

🔗 **Multiplayer Tetris** | C++, XMinig Server

- Tetris game using **OOP** principles to implement the **MVC** architecture through graphical/text display observers